

Sandwich-like Heterochromophore Metallo-Supramolecules Based on Dense Chromophore Arrangements with Energy and Chirality Transfer Properties

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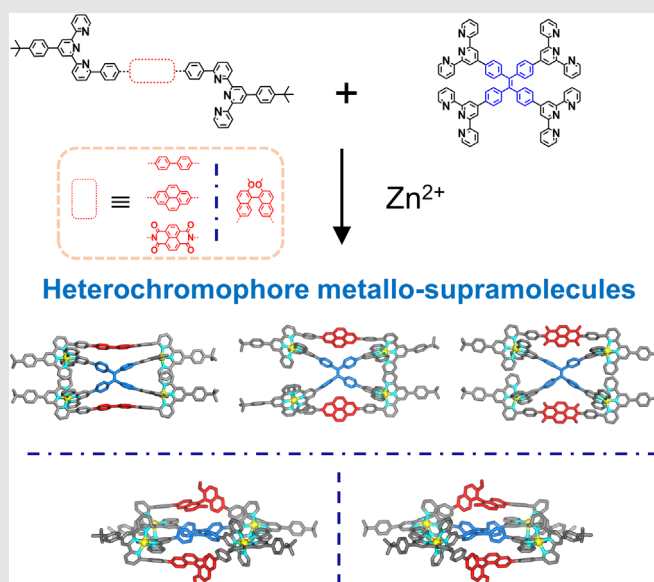
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Multichromophoric architectures with well-organized chromophoric arrangements are of vital importance to gain insights into interchromophoric interactions. Herein, we report a novel heteroleptic self-assembly strategy and construct three achiral and one chiral sandwich-like heterochromophore complexes ($\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A-C}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$). Possible dynamic products, including homoleptic oligomers and isomers, are eliminated, producing achiral complexes with exceptional purity. Isomers are observed in chiral complexes due to the flexible geometry of L^{D} . Perfectly aligned chromophore arrangements with precise spacing and orientations are evidenced by single-crystal X-ray diffraction. The distinct segments in the ligands lead to different interchromophoric interactions. Impressively, energy transfer and tunable emission, including white-light emission, are achieved for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$. In addition, chirality transfer is observed for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$, due to conformational restriction from chiral L^{D} , leading to chiral rotational conformations of tetraphenylethylene moiety in L^{G} .



Keywords: heterochromophore complexes, fluorescence, single crystal, dissymmetrical ligands, chirality

Introduction

Multichromophoric nanostructures^{1–3} provide an effective alternative for enabling efficient and tunable light-emitting or light-harvesting properties, deriving from improved control over the type, relative ratio, and positioning of chromophores compared to single-chromophore systems. Moreover, these systems can serve as ideal models for the investigation of interchromophoric interactions and photophysical pathways, such as electron transport,^{4–6} excitation energy transfer,^{7–9} chirality transfer,^{10–12} and through-space charge transfer.^{13–17} Thus, efficient and general construction strategies for fabricating multichromophoric architectures have attracted continuous attention. Covalent methods for preparing discrete multichromophoric structures, such as “V”- or “U”-shaped π -stack molecules^{17–19} and cyclophanes,^{10,20,21} are hampered by tedious purification steps and low yields. Noncovalent self-assembly can be an effective

approach to the synthesis of various architectures with high complexity. The difficulty of suppressing unexpected aggregation^{22,23} in solution, however, renders well-aligned chromophore assemblies challenging to fabricate through noncovalent interactions, especially for intricate systems containing multiple types and increased numbers of chromophores.²⁴

Heteroleptic metal-organic self-assembly^{25,26} has been widely applied to engineering nanostructures with multiple types of ligands incorporated. Statistical mixtures resulting from the low accuracy of self-assembly can be avoided through protected metal nodes,^{27–31} coordination sphere engineering,^{32,33} shape complementary assembly,^{34–36} and backbone-centred steric hindrance.³⁷ These strategies also allow for controllable stoichiometry of chromophores within different ligands.^{36,38–41} Abundant strategies reported to date of functional metallacycles and metallacages^{42,43} reveal a preference for a confined cavity. The scattered location of the chromophores

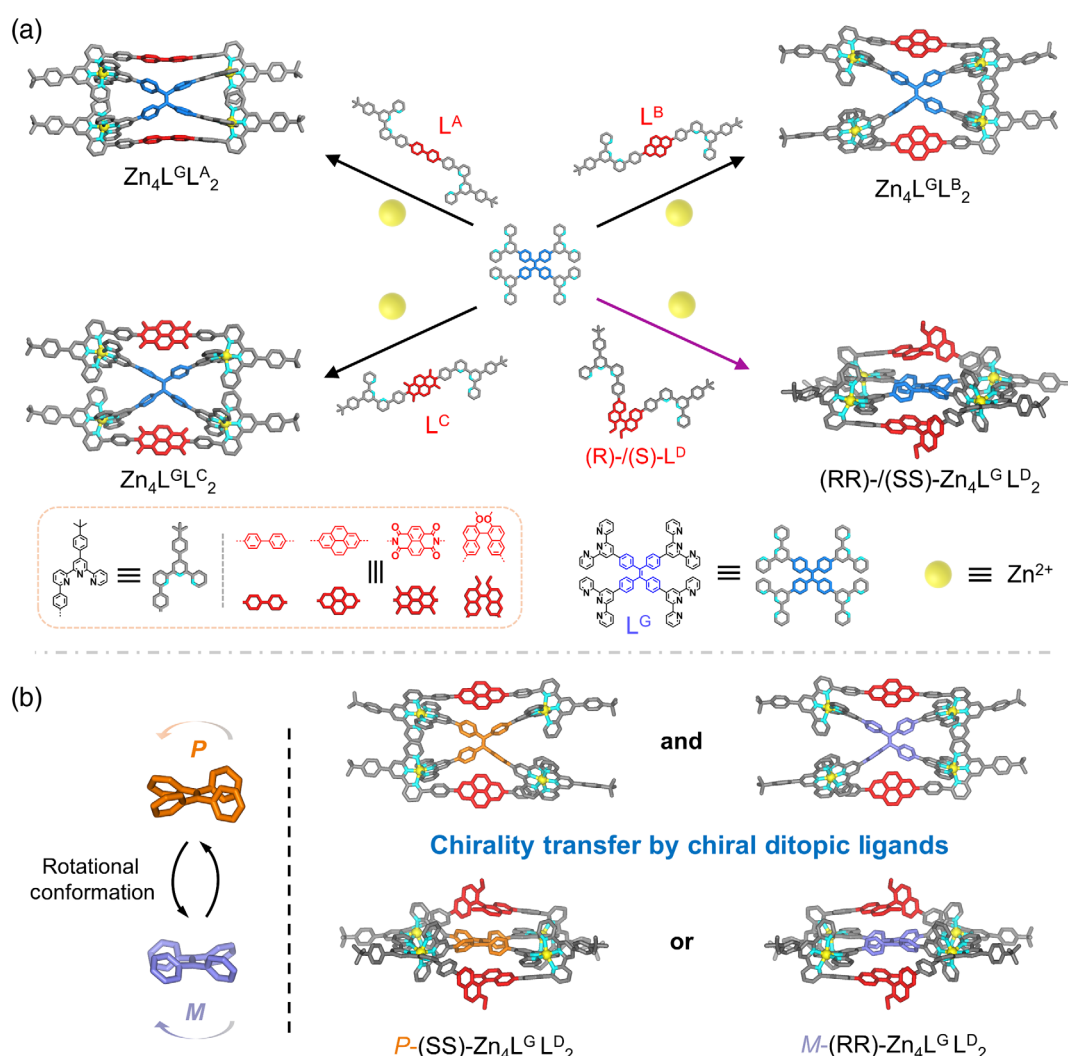


Figure 1 | (a) Design and self-assembly of ligands and heterochromophore three-component complexes. (b) Schematic illustration of chirality transfer in the complexes.

results in weakened through-space interactions and unexpanded functions. However, very few construction approaches reported to date can fulfill this demand.

As a tridentate donor ligand, 2,2':6',2''-terpyridine (tpy) has attracted much interest in the fields of supramolecular and coordination chemistry, due to its coordination ability with a variety of metal ions^{44–46} and the potential applications of tpy-based complexes such as catalysis,^{46–48} sensing,^{49,50} phototherapy,^{51–53} and so on. The stable connectivity of <tpy-M²⁺-tpy> has been widely used for the construction of metallo-supramolecules^{54–58} with high degrees of complexity. Due to the multiple sites that can be easily substituted within tpy units,⁵⁹ diverse chromophores can be widely introduced to realize impressive properties. In this work, we report a general strategy for preparing tpy-based heterochromophore complexes by employing shape complementary assembly (Figure 1a,b). The obtained three-component complexes include achiral Zn₄L^GL^A₂, Zn₄L^GL^B₂, Zn₄L^GL^C₂, and chiral (RR)-/(SS)-Zn₄L^GL^D₂. Ditopic achiral ligands (L^A, L^B, and L^C) with modified tpy motifs can accommodate tetratopic ligand L^G with high accuracy in the presence of Zn(NO₃)₂·6H₂O, whereas homoleptic coordination oligomers are absent. Single-crystal X-ray diffraction analysis confirms the configuration of achiral complexes with highly-aligned chromophore arrangements, rather than strained triple-decker structures. On the other hand, the flexible geometry of the chiral ditopic ligand L^D leads to the formation of triple-decker complex (RR)-/(SS)-Zn₄L^GL^D₂ with predominant *gem* configuration. The shortened distance between the ditopic ligand (namely L^H) and L^G enables efficient interchromophoric interactions, which alters the fluorescent behaviors. Benefiting from the fluorescent nature of L^H and L^G, tunable emission can be realized by changing solvent and concentration. For Zn₄L^GL^A₂ and Zn₄L^GL^B₂, white-light emission is achieved in dimethyl sulfoxide (DMSO) at a concentration of 40 μM (CIE chromaticity coordinate of *x* = 0.29, *y* = 0.32 for Zn₄L^GL^A₂ and *x* = 0.30, *y* = 0.35 for Zn₄L^GL^B₂). In addition, chirality transfer is observed in (RR)-/(SS)-Zn₄L^GL^D₂ due to the chiral rotational conformations of tetraphenylethylene (TPE) induced by the chiral binaphthyl core in ligand L^D.

Experimental Methods

All reagents were purchased from Sigma-Aldrich (Beijing, China), Matrix Scientific (Beijing, China), and Alfa Aesar (Shanghai, China) and used without further purification. 4'-Bromo-2,2':6',2''-terpyridine was purchased from Jilin Chinese Academy of Sciences-Yanshen Technology Co., Ltd. (Changchun, China). Fully synthetic procedures for all the ligands are described in the Supporting Information.

Nuclear magnetic resonance (NMR) spectra data were recorded on 400 MHz, 500 MHz, and 600 MHz Bruker

Advance NMR spectrometers (Bruker Switzerland AG, Fällanden, Zurich, Switzerland) in CDCl₃ or CD₃CN with tetramethylsilane (TMS) as reference. Electrospray ionization (ESI) and traveling-wave ion mobility (TWIM) mass spectra were recorded with a Waters Synapt G2 tandem mass spectrometer (Waters Corporation, Milford, Massachusetts, United States). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) was performed on a Bruker AutoflexIII (Bruker Corporation, Billerica, Massachusetts, United States) using trans-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyli-dene]malononitrile as a matrix. UV-vis spectra of solutions were recorded on a PerkinElmer LAMBDA 365 spectrophotometer (PerkinElmer, Shelton, Connecticut, United States). Fluorescence emission spectra were measured by a Shimadzu spectrofluorimeter RF-5301PC (Shimadzu, Kyoto, Japan). Absolute photoluminescence (PL) quantum yields and fluorescence lifetimes were carried out with FLS 980 spectrometer (Edinburgh Instruments Ltd., Livingston, United Kingdom). Absolute PL quantum yield was measured via an integrating light sphere. CD spectra were recorded using Chirascan-Plus CD (Applied Photophysics, Leatherhead, United Kingdom) spectrometers. Circularly polarized luminescence (CPL) spectra were recorded using JASCO CPL-300 spectrofluorescence polarimeter (Tokyo, Japan). X-ray diffraction data for Zn₄L^GL^B₂, Zn₄L^GL^C₂, and (RR)-Zn₄L^GL^D₂ were collected using synchrotron radiation and a MAR325 charge-coupled detector at Shanghai Synchrotron Radiation BL17B Beamline.

Synthesis of complex Zn₄L^GL^A₂

To a solution of ligand L^A (3.0 μmol, 3.1 mg) and ligand L^G (1.5 μmol, 1.9 mg) in CHCl₃ (6.0 mL), a solution of Zn(NO₃)₂·6H₂O (6.0 μmol, 1.8 mg) in MeOH (12.0 mL) was added, and then the mixture was kept in 50 °C for 10 h. After cooling to room temperature, excess NH₄PF₆ (around 100 mg) was added to generate a yellow precipitate (6.2 mg, 87%). ¹H NMR (600 MHz, CD₃CN) δ 9.11 (d, *J* = 1.4 Hz, 1H, tpy-*H*^{e'}), 8.88–8.83 (m, 2H, tpy-*H*^e and tpy-*H*^z), 8.67–8.59 (m, 4H, tpy-*H*^x_e and tpy-*H*^x₃), 8.34 (dd, *J* = 8.0, 7.2 Hz, 1H, tpy-*H*^{d'}), 8.30 (d, *J* = 8.0 Hz, 1H, tpy-*H*^z), 8.18 (d, *J* = 8.1 Hz, 2H, Ph-*H*^b), 8.13–8.10 (m, 2H, tpy-*H*^x₄), 7.85 (d, *J* = 5.2 Hz, 2H, tpy-*H*^x₆), 7.83–7.80 (m, 4H, Ph-*H*^a and Ph-*H*^d), 7.61 (d, *J* = 7.9 Hz, 2H, Ph-*H*^g), 7.55–7.52 (m, 3H, tpy-*H*^f and Ph-*H*^h), 7.48 (d, *J* = 8.1 Hz, 2H, Ph-*H*^c), 7.45 (ddd, *J* = 7.2, 5.2, 1.0 Hz, 2H, tpy-*H*^x₅), 7.13–7.10 (m, 3H, Ph-*H*^x₉ and tpy-*H*^d), 6.73 (dt, *J* = 5.2, 1.4 Hz, 1H, tpy-*H*^e), 6.46 (d, *J* = 7.7 Hz, 2H, Ph-*H*^x_h), 6.23 (ddd, *J* = 6.9, 5.4, 1.0 Hz, 1H, tpy-*H*^f), 1.47 (s, 9H, *H*^f). ¹³C NMR (125 MHz, CD₃CN) δ 161.61, 156.98, 156.46, 155.97, 155.67, 151.87, 150.82, 150.09, 149.37, 149.23, 149.05, 148.62, 148.30, 147.43, 146.20, 143.89, 142.34, 142.29, 141.68, 140.66, 140.47, 138.83, 138.08, 135.53, 134.06, 132.81, 128.85, 128.64, 128.52, 128.21, 127.65, 127.46, 127.15, 126.88, 124.35, 123.68,

123.62, 122.63, 122.33, 121.51, 35.69, 31.37. ESI-MS (m/z): 1039.3 $[M-4PF_6]^{4+}$ (calcd m/z : 1039.3), 802.4 $[M-5PF_6]^{5+}$ (calcd m/z : 802.4), 644.5 $[M-6PF_6]^{6+}$ (calcd m/z : 644.5), 531.7 $[M-7PF_6]^{7+}$ (calcd m/z : 531.7), 447.1 $[M-8PF_6]^{8+}$ (calcd m/z : 447.1).

Synthesis of complex $Zn_4L^GL^B_2$

To a solution of ligand L^B (4.2 μ mol, 4.4 mg) and ligand L^G (2.1 μ mol, 2.6 mg) in $CHCl_3$ (5.0 mL), a solution of $Zn(NO_3)_2 \cdot 6H_2O$ (8.4 μ mol, 2.5 mg) in MeOH (10.0 mL) was added, and then the mixture was kept in 50 °C for 10 h. After cooling to room temperature, excess NH_4PF_6 (around 100 mg) was added to generate a yellow precipitate (8.5 mg, 84%). 1H NMR (600 MHz, CD_3CN) δ 9.13 (d, J = 1.4 Hz, 1H, tpy- H^e), 8.88 (dd, J = 8.1, 0.9 Hz, 1H, tpy- H^b), 8.85 (d, J = 1.4 Hz, 1H, tpy- H^e), 8.68–8.64 (m, 4H, tpy- H^{xe} and tpy- H^{x3}), 8.37 (dd, J = 8.1, 7.1 Hz, 1H, tpy- H^d), 8.28 (d, J = 8.1 Hz, 1H, tpy- H^b), 8.22 (s, 2H, H^g), 8.20–8.17 (m, 4H, tpy- H^{x4} and Ph- H^b), 8.05 (d, J = 1.0 Hz, 2H, H^h), 7.89 (d, J = 4.5 Hz, 2H, tpy- H^{x6}), 7.83 (d, J = 8.0 Hz, 2H, Ph- H^a), 7.74 (d, 2H, J = 8.1 Hz, Ph- H^d), 7.58 (d, J = 6.8 Hz, 1H, tpy- H^f), 7.49 (ddd, J = 7.2, 5.2, 1.0 Hz, 2H, tpy- H^{x5}), 7.38 (d, J = 7.6 Hz, 2H, Ph- H^{xg}), 7.26 (d, J = 8.0 Hz, 2H, Ph- H^e), 7.07 (ddd, J = 8.4, 7.0, 1.7 Hz, 1H, tpy- H^d), 6.66 (dt, J = 5.3, 1.3 Hz, 1H, tpy- H^e), 6.55–6.52 (m, 2H, Ph- H^{xh}), 6.16 (ddd, J = 6.7, 5.4, 1.0 Hz, 1H, tpy- H^f), 1.48 (s, 9H, H^f). ^{13}C NMR (125 MHz, CD_3CN) δ 161.55, 156.98, 155.95, 155.87, 151.85, 150.78, 150.08, 149.47, 149.32, 149.07, 148.57, 148.38, 147.41, 145.39, 143.42, 142.63, 142.42, 142.31, 140.54, 138.10, 137.70, 135.35, 134.05, 132.35, 132.19, 128.93, 128.84, 128.77, 128.63, 128.54, 127.99, 127.63, 127.35, 126.21, 124.69, 124.45, 124.15, 123.63, 122.61, 122.32, 121.65, 35.66, 31.35. ESI-MS (m/z): 1063.3 $[M-4PF_6]^{4+}$ (calcd m/z : 1063.3), 821.6 $[M-5PF_6]^{5+}$ (calcd m/z : 821.6), 660.5 $[M-6PF_6]^{6+}$ (calcd m/z : 660.5), 545.4 $[M-7PF_6]^{7+}$ (calcd m/z : 545.4), 459.1 $[M-8PF_6]^{8+}$ (calcd m/z : 459.1).

Synthesis of complex $Zn_4L^GL^C_2$

To a solution of ligand L^C (3.0 μ mol, 3.4 mg) and ligand L^G (1.5 μ mol, 1.9 mg) in $CHCl_3$ (5.0 mL), a solution of $Zn(NO_3)_2 \cdot 6H_2O$ (6.0 μ mol, 1.8 mg) in MeOH (10.0 mL) was added, and then the mixture was kept in 50 °C for 10 h. After cooling to room temperature, excess NH_4PF_6 (around 100 mg) was added to generate a yellow precipitate (6.1 mg, 82%). 1H NMR (600 MHz, CD_3CN) δ 9.14 (s, 1H, tpy- H^e), 8.92 (d, J = 8.0 Hz, 1H, tpy- H^b), 8.87–8.81 (m, 3H, tpy- H^{xe} and tpy- H^e), 8.62 (d, J = 8.0 Hz, 2H, tpy- H^{x3}), 8.54 (s, 2H, H^g), 8.41 (t, J = 7.6 Hz, 1H, tpy- H^d), 8.28 (d, J = 8.1 Hz, 1H, tpy- H^b), 8.21–8.15 (m, 6H, Ph- H^d , tpy- H^{x4} and Ph- H^b), 7.87–7.82 (m, 4H, tpy- H^{x6} and Ph- H^a), 7.67–7.63 (m, 3H, tpy- H^f and Ph- H^e), 7.50–7.46 (m, 2H, tpy- H^{x5}), 6.99 (t, J = 7.2 Hz, 1H, tpy- H^d), 6.87 (d, J = 7.6 Hz, 2H, Ph- H^{xg}), 6.69 (d, J = 7.6 Hz, 2H, Ph- H^{xh}), 6.59 (d, J = 5.0 Hz, 1H, tpy- H^e), 6.30–6.26 (m, 1H, tpy- H^f), 1.48 (s, 9H, H^f). ^{13}C NMR

(125 MHz, CD_3CN) δ 163.47, 161.13, 156.99, 156.03, 155.81, 152.03, 150.85, 150.63, 149.82, 149.10, 148.72, 148.53, 147.35, 145.97, 143.58, 142.50, 142.41, 140.44, 138.77, 137.48, 136.06, 134.06, 132.44, 131.45, 129.69, 129.40, 129.07, 128.91, 128.55, 127.95, 127.82, 127.67, 127.55, 127.30, 124.33, 124.00, 123.70, 122.75, 122.45, 118.26, 35.72, 31.40. ESI-MS (m/z): 1095.2 $[M-4PF_6]^{4+}$ (calcd m/z : 1095.2), 847.2 $[M-5PF_6]^{5+}$ (calcd m/z : 847.2), 681.8 $[M-6PF_6]^{6+}$ (calcd m/z : 681.8), 563.7 $[M-7PF_6]^{7+}$ (calcd m/z : 563.7), 475.1 $[M-8PF_6]^{8+}$ (calcd m/z : 475.1).

Synthesis of complex (RR)-/(SS)- $Zn_4L^GL^D_2$

To a solution of ligand (R)-/(S)- L^D (3.0 μ mol, 3.6 mg) and ligand L^G (1.5 μ mol, 1.9 mg) in $CHCl_3$ (5.0 mL), a solution of $Zn(NO_3)_2 \cdot 6H_2O$ (6.0 μ mol, 1.8 mg) in MeOH (10.0 mL) was added, and then the mixture was kept in 50 °C for 10 h. After cooling to room temperature, excess NH_4PF_6 (around 100 mg) was added to generate a yellow precipitate (6.4 mg, 85%). ESI-MS (m/z): 1119.3 $[M-4PF_6]^{4+}$ (calcd m/z : 1119.3), 866.4 $[M-5PF_6]^{5+}$ (calcd m/z : 866.4), 697.9 $[M-6PF_6]^{6+}$ (calcd m/z : 697.9), 577.5 $[M-7PF_6]^{7+}$ (calcd m/z : 577.5), 487.2 $[M-8PF_6]^{8+}$ (calcd m/z : 487.2).

Results and Discussion

Synthesis and characterization of achiral complexes

In the synthesis of heterochromophore complexes, the elimination of homoleptic complexes posed a challenge. First, we designed and synthesized the tetratopic ligand L^G with a TPE core and four para-positioned tpy units which could afford two ditopic ligands with another chromophore. The homoleptic self-assembly of L^G in the presence of metal ions led to the formation of oligomeric complexes. The different geometric angles between the binding sites within L^G provided a clear design inspiration for nonstatistical complexes through shape complementary assembly. Moreover, the requirement to reduce the distance between chromophore segments should also be satisfied to realize effective interchromophoric interactions. The six-position modification⁶⁰ of tpy was a promising approach to reach these targets. The ligands incorporated with this hook-like unit can capture the ligands with matching shapes following the addition of metal ions. Accordingly, we designed and synthesized the ditopic ligands L^A , L^B , and L^C with similar arm lengths and geometric angles of 180°. Considering the linear configuration of the geometric angles and <tpy- M^{2+} -tpy> connectivity,^{61,62} homoleptic complexes of the ditopic ligands would be one-dimensional coordination oligomers. All ligands were synthesized in decent yields through the synthetic procedures in the Supporting Information Scheme S1 and fully characterized by 1H and ^{13}C NMR spectroscopy, two-dimensional correlation

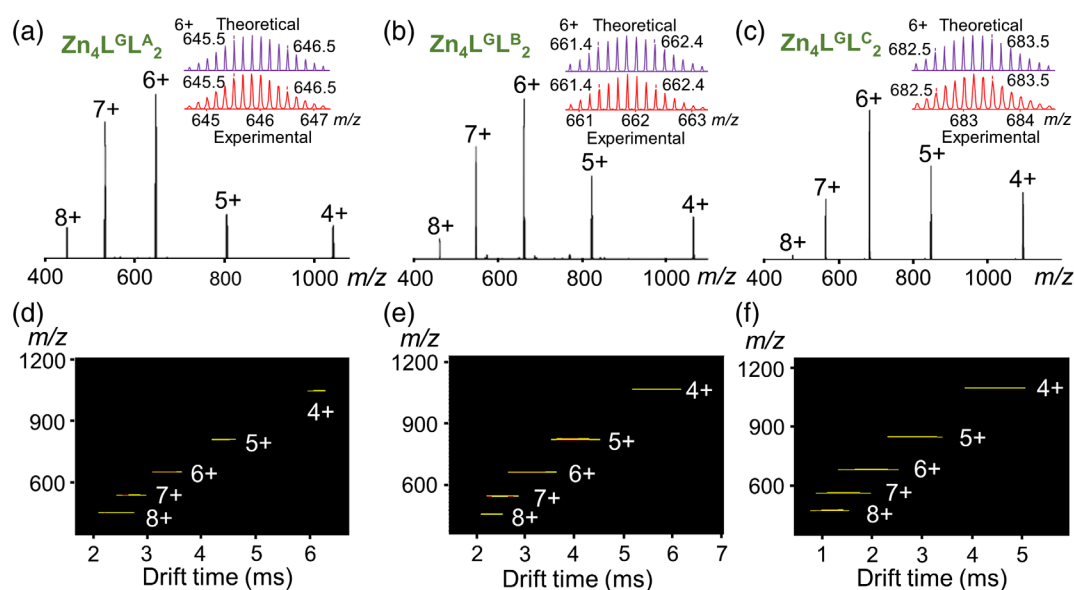


Figure 2 | ESI-MS and TWIM-MS spectra of (a) and (d) $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$, (b) and (e) $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, and (c) and (f) $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$.

spectroscopy (2D-COSY), and MALDI-TOF-MS (Supporting Information Figures S4–S26).

Heterochromophore three-component complexes were successfully prepared by mixing the ditopic ligand $\text{L}^{\text{A-C}}$, the tetratopic ligand L^{G} , and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in exact stoichiometric ratios of 2:1:4 in $\text{CHCl}_3/\text{MeOH}$ at 50 °C for 10 h. Following excess addition of NH_4PF_6 , yellow complexes were obtained without additional purification procedures. Electrospray ionization mass spectrometry (ESI-MS) displayed a set of prominent peaks arising from consecutive charge states (8+ to 4+) due to the loss of anions (PF_6^-), providing evidence for the expected formation of the heteroleptic complexes $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ (Figure 2a–c). In addition, the experimental isotope patterns of each charge state closely matched the corresponding simulated patterns (Supporting Information Figures S1–S3). To investigate the possible existence of overlapping isomers or conformers, TWIM-MS was employed (Figure 2d–f). The narrow drift time distributions of each charge state for these complexes indicated the formation of a single product.⁶³

Additionally, NMR spectroscopy was performed to confirm the generation of these complexes. ^1H NMR spectra exhibited sharp and well-resolved proton signals, providing another strong indication of the formation of discrete assemblies with high symmetry (Figure 3a–c). All NMR resonances of complexes were further assigned by 2D-COSY and nuclear Overhauser effect spectroscopy (NOESY) (Supporting Information Figures S27–S44). Through accurate integration analysis, the stoichiometries of these complexes were proved to be $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$, consistent with the expected complexes. The six-position modification within ditopic ligands altered the chemical environment of protons in tpy units, resulting in

signal splitting in the spectra of the ditopic ligands and complexes. The proton signals of 6-tpy in the ditopic ligands shifted upfield ($\Delta\delta \approx 2.08$ ppm for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$, $\Delta\delta \approx 2.16$ ppm for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, and $\Delta\delta \approx 2.21$ ppm for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$) due to strong electron-shielding effect.^{64,65} Diffusion-ordered NMR spectroscopy (DOSY) showed that all the protons of each complex had the same band (Supporting Information Figures S45–S47), consistent with discrete assemblies with a single component. $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ exhibited similar diffusion coefficient (D) of $3.57 \times 10^{-10} \text{ m}^2\text{s}^{-1}$, $3.65 \times 10^{-10} \text{ m}^2\text{s}^{-1}$, and $3.48 \times 10^{-10} \text{ m}^2\text{s}^{-1}$, respectively.

Through the above structural characterization, heterochromophore three-component complexes were evidenced to be the exclusive product, rather than the homoleptic coordination oligomers. Benefiting from our smart strategy, the modified tpy motif within the ditopic ligands exhibited preferential binding to unmodified tpy units within L^{G} . Owing to the multiple types of binding sites present in L^{G} , including *Z*-oriented sites with a 180° angle, *E*-oriented sites with a 60° angle, and *gem*-oriented sites with a 120° angle,⁶⁶ the configuration of the complexes remained uncertain, on account of the possible isomers. *Z*-oriented sites had matching geometric angles in terms of linear configuration of the ditopic ligands. However, considering the arm lengths, the best-matching choice should be *gem*-oriented sites. Although both orientations guarantee the maximum site occupancy, the exclusive presence of a single configuration indicates that the stability of the final product is markedly superior to that of the other.

To further verify the influence of the chromophore nature on self-assembly, we investigated the self-assembly of Zn^{2+} with three different components,

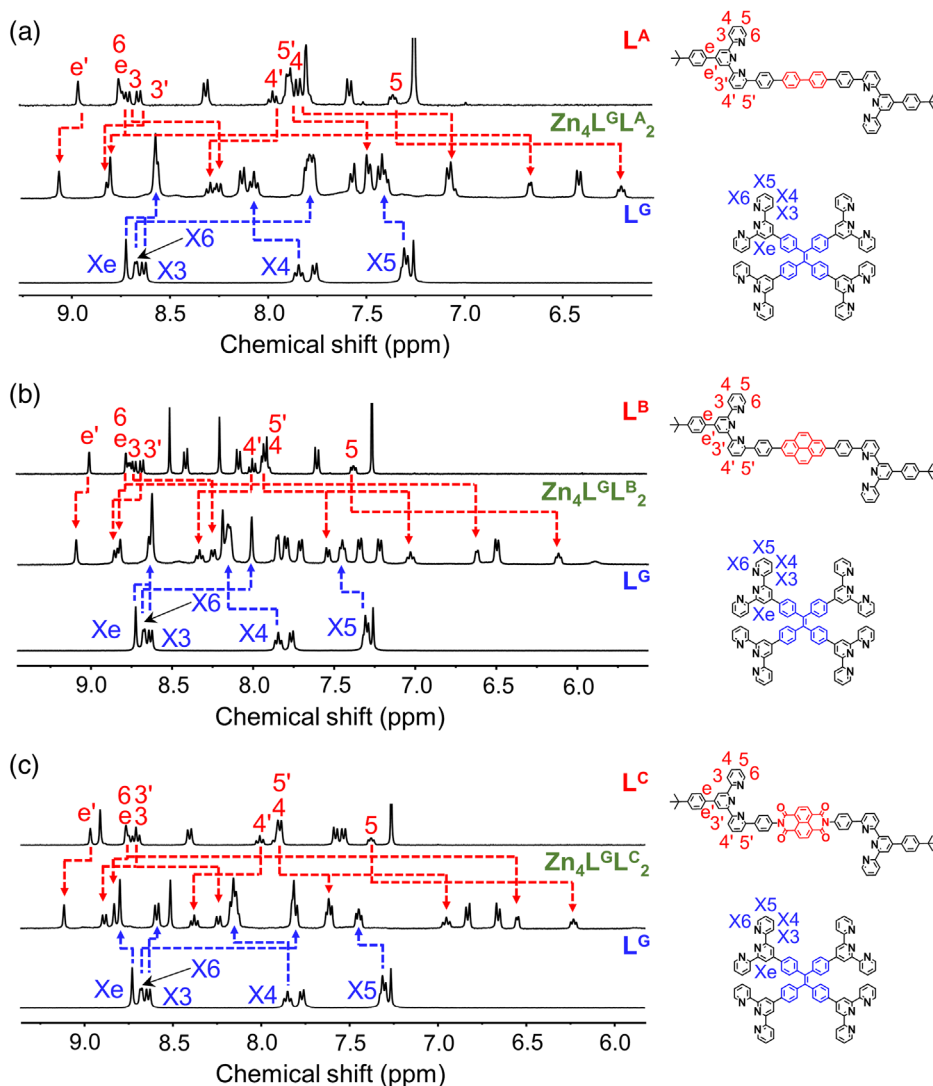


Figure 3 | ^1H NMR spectra (400 MHz, 298 K) of (a) ligand L^A , L^G in CDCl_3 , and complex $\text{Zn}_4\text{L}^G\text{L}^A_2$ in CD_3CN , (b) ligand L^B , L^G in CDCl_3 , and complex $\text{Zn}_4\text{L}^G\text{L}^B_2$ in CD_3CN , (c) ligand L^C , L^G in CDCl_3 , and complex $\text{Zn}_4\text{L}^G\text{L}^C_2$ in CD_3CN .

including two distinct ditopic ligands and L^G at 50 °C in $\text{CHCl}_3/\text{MeOH}$ for 10 h. After excess addition of NH_4PF_6 , yellow precipitates were obtained. The ^1H NMR spectra showed a set of new signals in all cases except for self-sorting products (Supporting Information Figures S48–S53). However, the complex signals made the exact composition challenging to determine. ESI-MS confirmed the occurrence of a statistically discrete mixture, including two self-sorting products and new complex with three different components (Supporting Information Figures S54–S59). Despite the different core segments, their identical shape resulted in the absence of additional driving forces for further increasing the binding selectivity. This result also indicates that our strategy for heterochromophore architectures mainly depends on the geometric shape, which shows promise for extensive introduction of diverse functional groups.

The configurations of complexes were further confirmed by single-crystal X-ray diffraction using synchrotron radiation (Figure 4a,b and Supporting Information Figures S90 and S91 and Tables S5 and S6). Single crystals were obtained under slow diffusion of 1,4-dioxane into a $\text{Zn}_4\text{L}^G\text{L}^B_2$ solution in $\text{CH}_3\text{CN}/\text{CH}_3\text{NO}_2$ and of isopropyl ether into a $\text{Zn}_4\text{L}^G\text{L}^C_2$ solution in $\text{CH}_3\text{CN}/\text{DMF}$ (N,N-dimethylformamide). Although the collection of sufficient diffraction data for $\text{Zn}_4\text{L}^G\text{L}^A_2$ was unsuccessful due to its weakly diffracting nature, identical configurations can be assumed for three complexes, as discussed above. Notably, the ditopic ligands L^H selectively bound in *gem*-oriented sites rather than *Z*-oriented sites. To accommodate L^H with incompatible angles, the metal center adopted a strained coordination geometry, with a small torsion angle of 21° for $\text{Zn}_4\text{L}^G\text{L}^B_2$ and 23° for $\text{Zn}_4\text{L}^G\text{L}^C_2$. The tpy units in L^G rotated and maintained a fixed configuration nearly perpendicular to the TPE plane. Because the

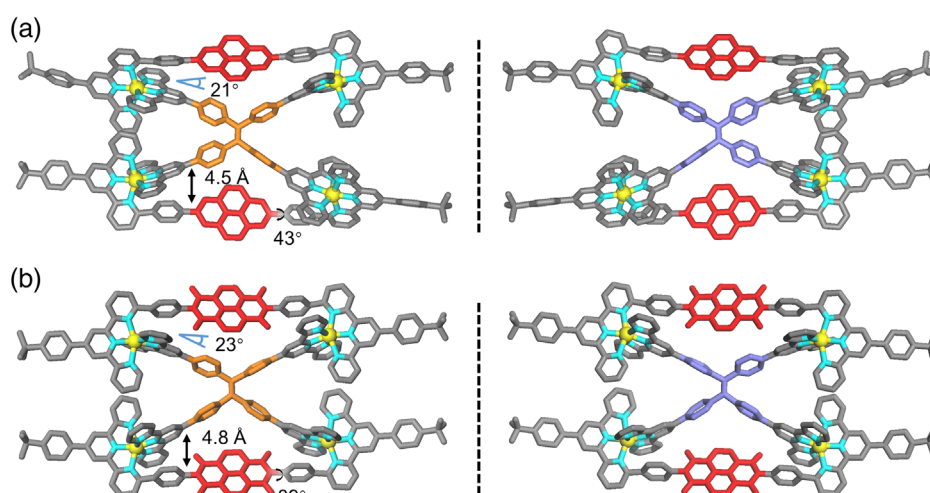


Figure 4 | X-ray crystal structures of enantiomeric complexes (a) $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ and (b) $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$. All H atoms, solvent molecules, and counterions are omitted for clarity.

pyrene and naphthalene diimide (NDI) cores were shorter than the TPE unit measured along the Z orientation, we inferred that flexible L^{G} would suffer from severe bending as a result of the higher strain in $\text{Z-Zn}_4\text{L}^{\text{G}}\text{L}^{\text{H}}_2$. To reduce the steric barrier associated with the hydrogen atoms of the adjacent benzene in L^{B} and L^{C} , the pyrene and NDI units exhibited large torsion angles of 43° and 89° , respectively. Two types of chromophores, TPE and pyrene or NDI, were positioned along one of the C_2 axes within complexes possessing D_{2h} symmetry. The close arrangement of the chromophores was also confirmed by the distances of 4.5 and 4.8 Å estimated for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$, respectively, based on the distance between single bonds attached to chromophores. Despite the insufficient cavity volumes for host-guest interactions, the highly ordered assignment and optimized distances between chromophores rendered it possible for stronger interchromophoric interactions.

Interestingly, these two complexes displayed different packing arrangements. A layer-by-layer configuration (Supporting Information Figure S85) was observed in the crystal structure of $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$. Additionally, PF_6^- counterions filled the gap between layers and formed abundant C-H...F hydrogen bonds with protons within $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ (Supporting Information Figure S86). Within each layer, the adjacent $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ exhibited loose packing manners through moderate π - π interactions between pyrene and tpy units. In comparison, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ was densely packed and showed multiple weak interactions including C-H... π interactions and π - π interactions as well as C-H...O hydrogen bonds (Supporting Information Figures S87 and S88). In addition, the enantiomers of both $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ were observed (Figure 4a, b), which was ascribed to the lack of restrictions on the chiral left-handed (M) and right-handed (P) rotational conformations (Figure 1b) of TPE moieties.^{67,68}

Tunable emission and energy transfer of achiral complexes

The photophysical properties of all the present ligands and complexes in solution were investigated by UV-vis absorption (Figure 5a,c and Supporting Information Figure S60) and fluorescence spectroscopies (Figure 5b, d). The absorption spectra of the four ligands at a concentration of 10 μM in CHCl_3 showed broad bands in the range of 250–360 nm, corresponding to the electronic transition of tpy and their respective chromophore units (Figure 5a). Ligands L^{B} and L^{C} displayed well-defined vibronic absorption bands in the ranges of 310–358 nm and 333–405 nm, corresponding to characteristic electronic transitions of pyrene and NDI units, respectively. The intense shoulder peak from 350 to 440 nm in the spectrum of L^{G} was ascribed to the intramolecular charge transfer (ICT) transition from TPE to tpy.

The PL spectra of all three L^{H} (Figure 5b) showed emission bands with maximum emission wavelengths around 385 nm for L^{A} , 422 nm for L^{B} , and 421 nm for L^{C} , overlapping with the absorption bands of L^{G} (Supporting Information Figure S61). Unlike the adequate fluorescence properties of L^{A} (absolute fluorescence quantum yield $\Phi_{\text{F}} = 96.7\%$) and L^{B} ($\Phi_{\text{F}} = 7.1\%$), L^{C} displayed negligible emission in CHCl_3 ($\Phi_{\text{F}} < 0.1\%$), originating from deactivation through close triplet states^{69,70} (Supporting Information Figure S62). L^{G} exhibited broad and featureless emission bands around 550 nm, corresponding to ICT transition (Figure 5b). The unrestricted free rotation of TPE units^{71–73} led to nonradiative decay to quench the fluorescence of L^{G} ($\Phi_{\text{F}} < 0.1\%$, Supporting Information Figure S62).

Compared to ligand L^{G} , both $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ exhibited dramatically enhanced Φ_{F} of 5.5% and 9.6% in CH_3CN , respectively (Figure 5d and Supporting

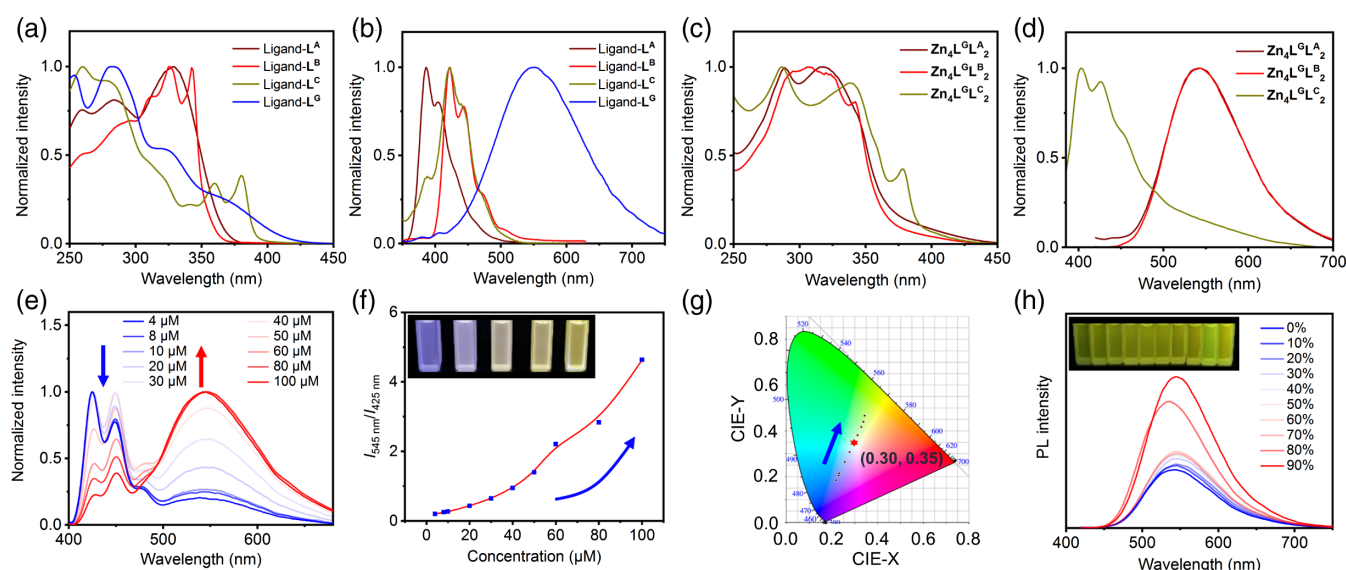


Figure 5 | (a) Absorption and (b) fluorescent emission spectra of ligands L^A , L^B , L^C , and L^G at the concentration of $10\ \mu\text{M}$ in CHCl_3 . (c) Absorption and (d) fluorescent emission spectra of complexes $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ at the concentration of $5\ \mu\text{M}$ in CH_3CN . (e) Fluorescent emission spectra ($\lambda_{\text{ex}} = 365\ \text{nm}$) of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ at different concentrations in DMSO. (f) Plot of relative fluorescent intensity ($I_{445\ \text{nm}}/I_{425\ \text{nm}}$) of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ versus concentration in DMSO. Inset: photograph of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ in DMSO at concentrations of 4 , 10 , 40 , 60 , and $100\ \mu\text{M}$ upon excitation at $365\ \text{nm}$ using a UV lamp at $298\ \text{K}$. (g) CIE coordinates of the emission of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ in DMSO at different concentration (4 – $100\ \mu\text{M}$). (h) Fluorescent emission spectra ($\lambda_{\text{ex}} = 365\ \text{nm}$) of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ versus the water fraction in $\text{CH}_3\text{CN}/\text{H}_2\text{O}$ at the concentration of $5\ \mu\text{M}$. Inset: photograph of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ in $\text{CH}_3\text{CN}/\text{H}_2\text{O}$ with different water fractions upon excitation at $365\ \text{nm}$ using a UV lamp at $298\ \text{K}$.

Information Figures S62 and S71 and Table S4). We inferred that coordination bonds and the steric effects of core segments in the ditopic ligands restricted intramolecular rotation and vibration to enable radiative decay. Although L^A and L^B showed excellent fluorescent properties in solution, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ exhibited low intensity in the emission regions of L^A and L^B . Considering the reduced distance and spectral overlap between L^H and L^G , L^G can quench the fluorescence from L^H through an efficient Förster resonance energy transfer process. Fluorescence titration measurements were carried out to monitor the process of self-assembly (Supporting Information Figure S63). With the increasing concentration of $\text{Zn}(\text{NO}_3)_2$ in the mixture solution of L^H and L^G in stoichiometric ratios of $2:1$, the fluorescence bands of L^A or L^B showed a downward trend, which could be rationalized by the energy transfer process from energy donor (L^A or L^B) to acceptor (L^G). The enhancement in the fluorescence band of L^G was due to the energy transfer and restricted intramolecular motion of TPE units. Eventually, the emission of L^A or L^B became negligible, indicating efficient energy transfer. In contrast, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ still showed weak emission in CH_3CN ($\Phi_{\text{F}} < 0.1\%$, Supporting Information Figure S62), which might be ascribed to the electron-deficient nature of NDI as a result of the electron transfer process.^{74–76}

Different solvents are known to have a marked impact on fluorescence properties. Fluorescence spectroscopy measurements were performed on the three complexes in several solvents (Supporting Information Figure S67). The two distinct fluorescence bands corresponding to L^H and L^G , with different intensity ratios, led to significant variations in fluorescent colors. This may be due to different association constants of complexes, energy transfer efficiencies from L^H to L^G , and the fluorescent nature of L^H and L^G in different solvents. The combination of blue and yellow emissions covered the entire visible range, enabling white-light emission through the modulation of the emissions of L^H and L^G . Benefiting from the dynamic nature of the coordination interaction and different emission behavior of L^H with aggregation-caused quenching (ACQ) and L^G with aggregation-induced emission (AIE) moieties, the intensity ratio of the two emission bands ($I_{\text{LG}}/I_{\text{LH}}$) was expected to depend on concentration. Both $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ exhibited weak emission from L^H and bright emission from L^G in CH_3CN ($1\ \mu\text{M}$). With the increase of concentration, $I_{\text{LG}}/I_{\text{LH}}$ ratio displayed a gradually upward trend (Supporting Information Figure S64), consistent with the measurements in DMSO (Figure 5e,f and Supporting Information Figure S65). We speculated that the ACQ effect of moieties in L^H and the AIE effect of the TPE moiety in L^G played an important role in this

phenomenon. Different from the yellow emission in CH_3CN , $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ showed tunable fluorescent colors from blue to yellow with the increasing concentration in DMSO. More importantly, both $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ exhibited white-light emission (CIE chromaticity coordinate of $x = 0.29$, $y = 0.32$ for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and of $x = 0.30$, $y = 0.35$ for $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$) at a concentration of 40 μM (Figure S5g and Supporting Information Figure S66 and Tables S1 and S2). In comparison to the weak emission in CH_3CN , the emission of $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ was considerably improved in DMSO, which suggested that the nonradiative pathway was highly dependent on solvent (Supporting Information Figure S65 and Table S3). However, reduced emission was observed at high concentration, similar to the ACQ effect.

Inspired by the incorporation of both AIE and ACQ fluorophores, the fluorescent properties of three complexes were further investigated in their aggregated states. The PL spectra were monitored in $\text{CH}_3\text{CN}/\text{water}$ mixtures with different water fractions (f_w). For $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$, with the increase of water fraction, the PL intensity increased slowly and reached the maximum at 90%, which was similar to AIE enhancement behavior of ligand L^{G} (Figure 5h and Supporting Information Figures S68 and S70). This result can be attributed to the enhanced restriction of intramolecular motion (RIM) of TPE segments after aggregation.^{77–79} Whereas $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ showed higher quantum yields ($\Phi_{\text{F}} = 57.7\%$) than $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ ($\Phi_{\text{F}} = 10.7\%$) in powder form (Supporting Information Figure S72). Different from the above two complexes, $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{C}}_2$ still displayed weak emission with increasing water fraction (Supporting Information Figure S69), consistent with its poor fluorescent properties in powder state ($\Phi_{\text{F}} < 0.1\%$, Supporting Information Figure S72). This behavior can be attributed to a nonradiative relaxation process resulting from strong interactions between NDI and TPE segments, despite the improved RIM.

Synthesis and characterization of chiral complexes

Encouraged by the successful synthesis of the three complexes, we expanded the scope of our strategy to the complexes with higher complexity and more functionalities. Enantiopure ditopic ligands (R)- and (S)- L^{D} were designed, synthesized, and characterized by ^1H and ^{13}C NMR spectroscopy, 2D-COSY, and MALDI-TOF-MS (Supporting Information Scheme S2 and Figures S73–S77). Benefiting from the axially chiral binaphthyl core between six-position-modified tpy units, the flexible geometric angles in (R)- or (S)- L^{D} allowed for the possible coordination modes with Z-, E-, and *gem*-oriented sites. Complex (RR)-/(SS)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ was synthesized by mixing the enantiopure ligand (R)-/(S)- L^{D} , ligand L^{G} , and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in exact stoichiometric ratios of 2:1:4 in $\text{CHCl}_3/\text{MeOH}$ at 50 $^\circ\text{C}$ for 10 h. After excess addition of

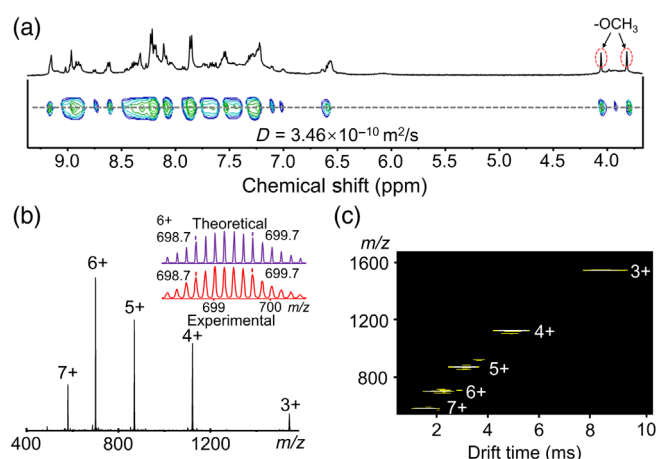


Figure 6 | (a) ^1H NMR and DOSY NMR spectra of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ in CD_3CN (500 MHz, 298 K). (b) ESI-MS and (c) TWIM-MS spectra of complex $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$.

NH_4PF_6 , yellow complexes were obtained without additional purification procedures.

ESI-MS and TWIM-MS evidenced the formation of expected three-component complex (RR)-/(SS)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ (Figure 6b,c and Supporting Information Figure S79). However, the aromatic region of the ^1H NMR spectrum displayed more complex signals (Supporting Information Figure S78) compared to above three achiral complexes. The signals of methoxyl groups within the binaphthyl core provide evidence for structural analysis. Notably, two main peaks and multiple lower peaks were observed, indicative of the different chemical environment resulting from isomers. The DOSY spectrum showed comparable diffusion coefficients ($D = 3.46 \times 10^{-10} \text{ m}^2\text{s}^{-1}$), reflecting the similar hydrodynamic radii of three types of $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ isomers (Figure 6a).

X-ray quality crystals were obtained by slow vapor diffusion of ethyl acetate into a mixture solution (CH_3CN and DMF) of (RR)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$. In the crystal structure (Figure 7a and Supporting Information Figures S89 and S92 and Table S7), only the C_2 -symmetric configuration where *gem*-oriented sites of L^{G} were selected by (R)- L^{D} was observed. However, the torsion angle between two naphthyl groups within the binaphthyl core ($\sim 88^\circ$) was smaller than that (120°) in *gem*-oriented sites. Apparently, the methoxyl groups within the same binaphthyl core were in different chemical environments, consistent with the signals in the ^1H NMR spectrum. The dense and ordered arrangement of chromophores was confirmed with a distance of $\sim 4.1 \text{ \AA}$, based on the distance between the single bond of the binaphthyl core and the double bond of the TPE core.

To assess the impact of the incorporated chiral core in the ligand L^{D} , chiroptical spectra of ligands (R)- and (S)- L^{D} and of complexes (RR)- and (SS)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ were further measured. The expected mirror-imaged circular

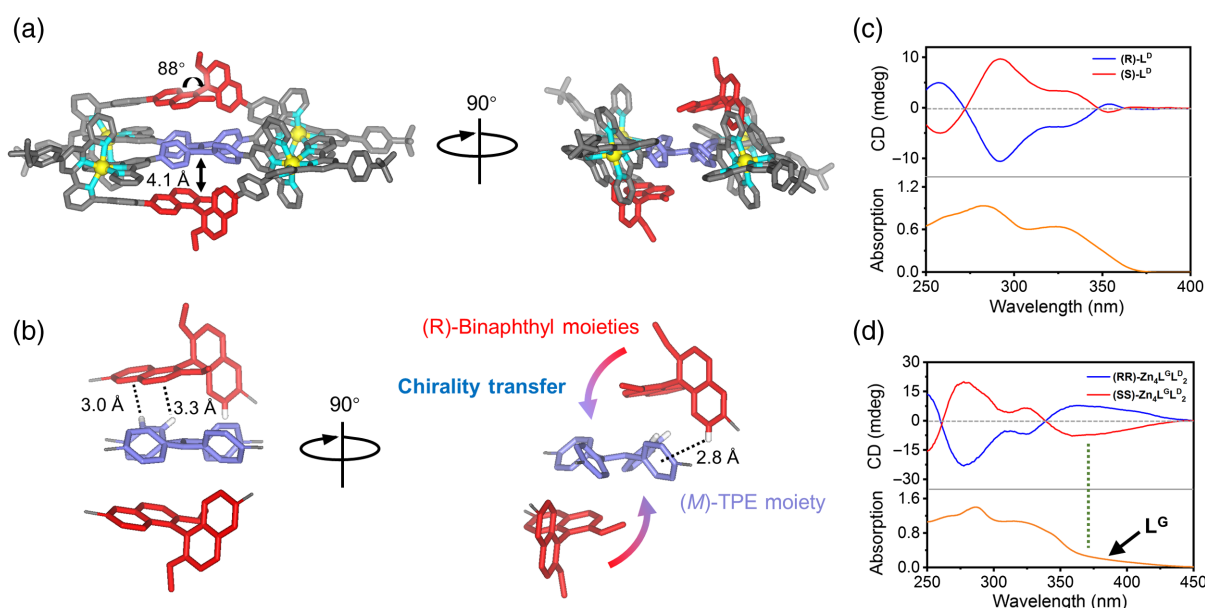


Figure 7 | (a) The entire view of the X-ray crystal structure of complex $(RR)\text{-Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ in different perspectives. (b) Binaphthyl and TPE moieties within the X-ray crystal structure of complex $(RR)\text{-Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ and C-H... π interactions (black dotted line) in different perspectives. All solvent molecules, counterions, and some H atoms are omitted for clarity. (c) CD spectra ($c = 5 \mu\text{M}$) and absorption spectra ($c = 10 \mu\text{M}$) of L^{D} enantiomers in CHCl_3 . (d) CD spectra ($c = 5 \mu\text{M}$) and absorption spectra ($c = 5 \mu\text{M}$) of $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ enantiomers in CH_3CN .

dichroism (CD) signals were observed for both (R)-/(S)- L^{D} and (RR)-/(SS)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$. The CD spectrum of (RR)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ displayed negative Cotton effects at 286 and 324 nm (Figure 7d and Supporting Information Figure S82), attributed to chiral ligand (R)- L^{D} , according to the very similar shape to the CD spectrum of (R)- L^{D} (Figure 7c and Supporting Information Figures S80 and S81). Compared with ligand (R)- L^{D} , complex (RR)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ showed additional positive Cotton effects with higher absorption dissymmetry factors (g_{abs}) at the ICT transition of achiral ligand L^{G} (centered at ~ 385 nm, $|g_{\text{abs}}| = 1.09 \times 10^{-3}$) and CPL (Supporting Information Figure S84), indicating the chirality transfer from (R)- L^{D} to L^{G} (Supporting Information Figure S83). We speculated that the chirality of L^{G} could originate from the conformational restriction of the binaphthyl core in (R)- L^{D} . In the crystal structure of (RR)- $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$, only the *M* conformation of the TPE core was observed (Figure 7a). This conformation was realized by the C-H... π interactions between TPE and binaphthyl moieties (Figure 7b), benefiting from the dense intramolecular arrangements of the two types of chromophores in solution.

Conclusion

In summary, we proposed a novel strategy for efficiently engineering heterochromophore architectures with shortened interchromophoric distance. Four heteroleptic three-component complexes (achiral $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A-C}}_2$ and

chiral $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$), comprising two ditopic L^{H} and a tetra-topic L^{G} , with perfectly aligned chromophore arrangements were designed and synthesized through shape complementary assembly. Single-crystal X-ray diffraction analysis confirmed their sandwich-like configurations. The improved interchromophoric interactions led to different fluorescent properties of achiral complexes in both solution and aggregated states. In addition, owing to the different emission nature of ligands (with ACQ or AIE moieties) and interchromophoric interactions, the solvent and concentration played a significant role in tunable dual emission for three complexes. Notably, white-light emission was achieved for both $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{A}}_2$ and $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{B}}_2$ in DMSO at the concentration of 40 μM . The flexible geometry of L^{D} led to the formation of $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ isomers where *gem* configuration was the predominant component. Chirality transfer was observed in $\text{Zn}_4\text{L}^{\text{G}}\text{L}^{\text{D}}_2$ due to conformational restriction from chiral ligand L^{D} . This work provides a facile and general method for selectively and efficiently engineering heterochromophore architectures with precise and predicable chromophore orientations and shortened interchromophoric distance. We believe that a wide range of aromatic molecules, including polycyclic and nonpolycyclic aromatic hydrocarbons, aromatic diimides, and aromatic heterocycles, could be readily introduced into this system to explore through-space interactions between functional units. We anticipate that this approach will effectively advance the properties of materials in the fields of emission, catalysis, chirality, and optoelectronic functions.

Supporting Information

Supporting Information is available and includes the synthesis of the ligands and complexes, NMR spectra, MALDI-TOF spectra, ESI-MS data, single-crystal data, absorption spectra, fluorescent emission spectra, CD spectra, and CPL spectra. CCDC 2239407 $\text{Zn}_4\text{L}^{\text{B}}_2$, 2239409 $\text{Zn}_4\text{L}^{\text{C}}_2$, and 2239410 (RR)- $\text{Zn}_4\text{L}^{\text{D}}_2$ contain the supplementary crystallographic data for this paper. The data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/structures.

Conflict of Interest

The authors declare no conflict of interest.

Acknowledgments

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